Lab 5 - Student Project

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In this experiment, we analyze how a flow behaves as it moves through a CD nozzle. The CD nozzle was designed in SolidWorks based on constraints presented in the Appendix. We compare the flow behavior with the anticipated results based on analytical methods derived from isentropic flow theory. The experimental data were collected using a pitot-tube setup, then imported into MATLAB to compare with the analytical results, and plotted together to show the similarities and differences between the two. This experiment is applicable in many real-world situations due to the various applications of CD nozzles. Rocket engines, jet engines, supersonic wind tunnels, propulsion systems, and many other systems use CD nozzles to accelerate flow. Predicting pressure ratios, Mach number distributions, and mass flow rates is crucial for designing stable propulsion systems.

Nomenclature

p^*	=	Critical Static Pressure
p_o	=	Stagnation/Total Pressure
T^*	=	Critical Temperature
T_o	=	Stagnation/Total Temperature
V	=	Velocity
A	=	Nozzle Area
A*	=	Area of throat (M=1)
M	=	Mach Number
γ	=	Specific Heat of Air
ρ	=	Density
ρ_o	=	Stagnation/Total Density
R	=	Ideal Gas Constant
a	=	Speed of Sound
$\dot{m}$	=	Mass Flow Rate

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Introduction

The objective of this experiment is to analyze the behavior of a flow through a CD nozzle and compare the results with theoretical predictions based on isentropic flow theory. CD nozzles are designed with a converging nozzle section followed by a diverging section; the point where the converging and diverging sections meet is called the throat. The nozzle is specially designed to "choke" the flow at the throat, which means the Mach number is one in the throat. Unlike converging nozzles, CD nozzles accelerate subsonic flows to the throat and can reach supersonic speeds within the diverging section. The supersonic flow is determined by the ratio between the nozzle's exit area and throat area. "The expansion of a supersonic flow causes the static pressure and temperature to decrease from the throat to the exit, so the amount of the expansion also determines the exit pressure and temperature. The exit temperature determines the exit speed of sound, which determines the exit velocity. The exit velocity, pressure, and mass flow through the nozzle determine the amount of thrust produced by the nozzle" (MAE 352 Lab-4 Handout).

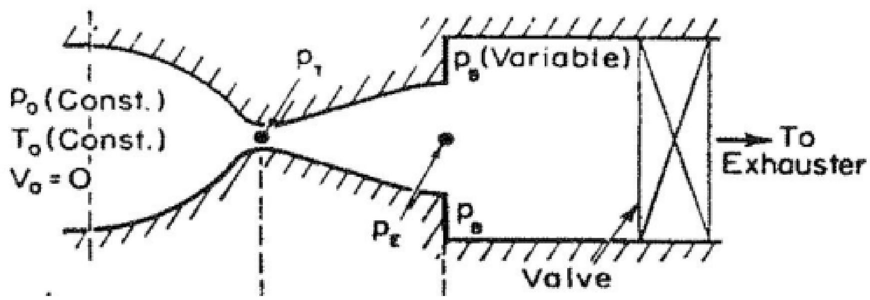

Figure 1: Schematic of a converging-diverging nozzle.

Figure 1 is from the MAE 352 Lab-4 Handout, which shows how a CD nozzle is set up.

$$\frac{p^*}{p_o} = 0.5283$$

$$\frac{T^*}{T_o} = 0.8333$$

Equations I-II

Equations 1 and 2 show the critical-condition ratios for both pressure and temperature, which can be used to solve for the stagnation pressure and temperature throughout the nozzle.

Once the flow is choked, three possible flow conditions can form in the diverging section.

1. Subsonic isentropic flow (the flow decelerates after the choked condition)
2. Supersonic non-isentropic flow (where the flow accelerates supersonically, forms a normal shock, and decelerates subsonically after the shock)
3. Supersonic isentropic flow (where the flow accelerates supersonically after the choked condition)

By observing the three conditions in the diverging section of the CD nozzle, seven profiles can be found:

1. Subsonic flow (never reaches choked condition).
2. Subsonic flow reaching choked condition, never reaching supersonic velocities (considered isentropic).
3. Subsonic flow reaching choked conditions produces a normal shock, followed by subsonic deceleration.
4. Subsonic flow reaching choked conditions produces a supersonic flow that forms a normal shock after the nozzle (assumed isentropic).
5. Over-expanded flow.

6. The flow after the choked condition is supersonic through the nozzle, so no shock forms.
7. Under-expanded flow.

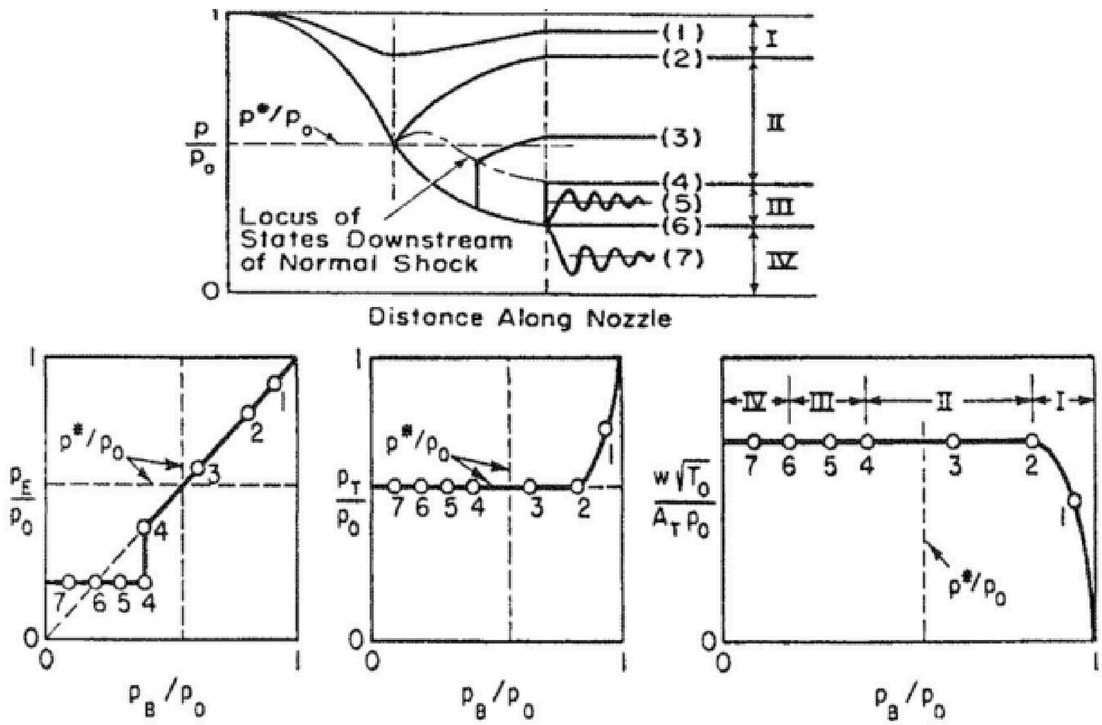

Figure 2: Flow conditions and regimes in a Converging-Diverging nozzle.

Methodology

The first step in this experiment is to calculate the expected results based on the theory analytically.

$$\frac{A}{A^*} = \left(\frac{\gamma+1}{2}\right)^{-\frac{\gamma+1}{2(\gamma-1)}} \frac{\left(1+\frac{\gamma-1}{2}M^2\right)^{\frac{\gamma+1}{2(\gamma-1)}}}{M}$$

$$\frac{p}{p_o} = \left(1+\frac{\gamma-1}{2}M^2\right)^{-\frac{\gamma}{\gamma-1}}$$

$$\frac{T}{T_o} = \left(1+\frac{\gamma-1}{2}M^2\right)^{-1}$$

$$\frac{\rho}{\rho_o} = \left(1+\frac{\gamma-1}{2}M^2\right)^{-\frac{1}{\gamma-1}}$$

$$a = \sqrt{\gamma RT} \qquad \text{(speed of sound)}$$

$$V = Ma$$

$$\dot{m} = \rho AV \qquad \text{(mass flow rate)}$$

Equations III-IX

Equations three through nine are the isentropic equations for calculating the properties of an isentropic flow. Using MATLAB, equation three can be solved numerically for the Mach number, which can then be used for pressure, density, temperature, velocity, and mass flow rate. The speed of sound is independent of the Mach number.

In designing our nozzle. We prioritized flow quality at the nozzle exit. This meant designing our nozzle to a particular tank pressure, which would yield a specific Mach number. In our case, we designed the nozzle to operate at 75 Psi, and given our ambient pressure of 14.7 Psi, we get a pressure ratio of 5.1. From isentropic tables, this corresponds to M = 1.72. At this M = 1.72, we can solve the Mach-area relation (eq.III) and an area ratio (A/A*) of 1.3586. To satisfy this, we reviewed our constraints. We opted for a throat radius of 0.2795 inches, which also prevents critical blockage of the duct by the pitot tube, since we ensured that the ⅛-inch-diameter instrument occupies only 5% of the throat area (0.2454 in^2)—this more than satisfies the 10% maximum. Lastly, to comply with the design area ratio, our exit radius needs to be 0.3258 inches (A = 0.3334 in^2). Below is the technical drawing of our nozzle design, followed by the SolidWorks model used for 3D printing. The nozzle design specifications and model are provided in Appendix I.

We experimentally tested our manufactured nozzle in NC State's wind tunnel facility. A flow is then driven by pressure using a compressed-air source. "The nozzle exit is open to the atmosphere, and a pressure regulator controls the total pressure in the tank. A Pitot tube installed on a traverse is used to measure the Mach number at the exit and in the nozzle" (MAE 352, Lab-3 Handout). Once the setup is complete, the pitot tube is aligned 1 inch outside the nozzle exit. The pressure is then increased by opening the pressure regulator to approximately 60 psi. Once the tank pressure is set to 60 psi by adjusting the pressure regulator, we begin collecting data. "The pitot tube is moved on the traverse inside the nozzle at discrete locations. The ball valve is used to shut off the compressed air in the tank when moving the pitot tube" (Handout).

The tests were conducted 15 times, once every half inch, until the pitot tube reached 5 inches to the throat, and then once every ⅕ inch until the pitot tube reached 6 inches. The data collected are the stagnation pressure and temperature in the reservoir, the stagnation pressure and temperature at the pitot tube, and the absolute pressure at the pitot tube. This data is collected and then imported into MATLAB for processing and manipulation for plotting.

In addition to the analytical and experimental data collected, a CFD analysis was conducted using ANSYS Fluent. This analysis was used to simulate the flow through the converging-diverging nozzle. The nozzle geometry was first generated in SolidWorks and imported into ANSYS Workbench. This geometry was then modified to extract the internal volume of the converging-diverging nozzle. The inlet, outlet, and wall surfaces were then defined within the meshing tool. After this preparation, an automatic mesh was generated over the surface and checked to ensure minimal artifacting. This meshed geometry was then imported into ANSYS Fluent, using air as the fluid and assuming inviscid, ideal conditions. The values for the pressurized tank, such as pressure and temperature, were assigned to the inlet, while atmospheric conditions were assigned at the outlet. For the solver, both the inlet and outlet were implemented as pressure-dependent. The solver was also configured to use a density solver rather than a pressure solver, as this produced more accurate results. These conditions were initialized from the inlet, and 687 iterations were run until a solution converged. At this convergence, the values for temperature, velocity, pressure, density, etc., were extracted and placed into an Excel sheet. Within the sheet, the data was polished to match local coordinates and normalized against nozzle length.

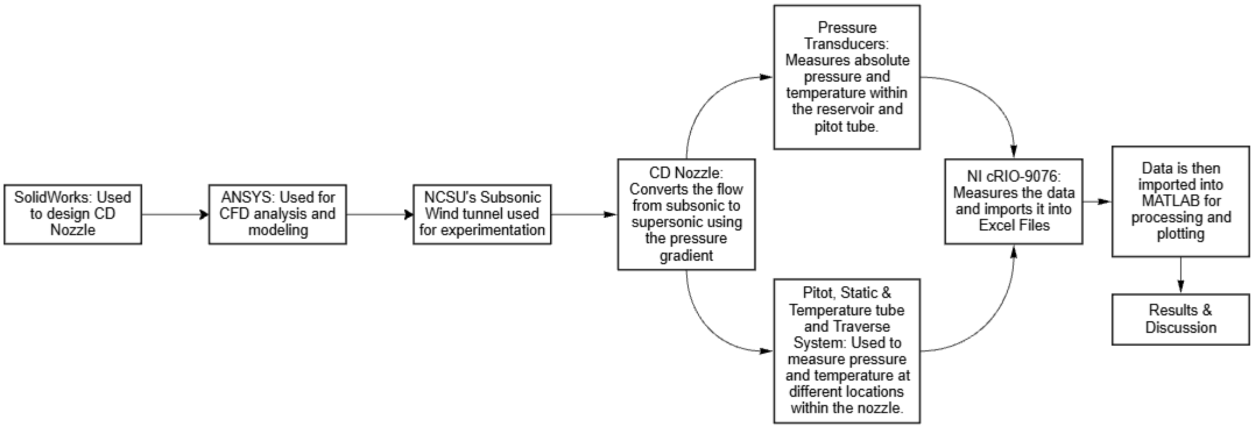

Figure 3: Experimental Flow Diagram

The experimental flow diagram shows the specific tools used to complete this experiment.

Results

The analytical isentropic relationships were used to determine the theoretical flow properties throughout the CD nozzle. Using the area–Mach number relation and the isentropic flow equations, the area, pressure, temperature, density, velocity, Mach number, and mass flow rate were calculated at 15 locations along the nozzle. The results are shown in Table 1.

R (J/K.kg)		γ		p_0 (Pa)		T_0 (K)		ρ_0 (kg/m^3)	
287		1.4		517,194		300		6.01	
Point	Location (mm)	Area (mm^2)	Pressure (Pa)	Temperature (K)	Density (kg/m^3)	Velocity (m/s)	Mach Number	Mass Flow (kg/s)	Note
1	0	490.82	504231	297.8	5.90	65.99	0.191	0.1911	Inlet
2	6.35	390.52	496368	296.5	5.83	83.88	0.243	0.1911	
3	12.7	301.67	481100	293.9	5.70	111.05	0.323	0.1911	
4	19.05	224.27	446386	287.6	5.41	157.56	0.464	0.1911	
5	25.4	158.32	273224	250.0	3.81	316.94	1.000	0.1911	Throat
6	38.1	163.61	210531	232.1	3.16	369.49	1.210	0.1911	
7	50.8	168.99	186020	224.0	2.89	390.77	1.303	0.1911	
8	63.5	174.45	168184	217.6	2.69	406.75	1.376	0.1911	
9	76.2	180.01	153916	212.2	2.53	420.01	1.438	0.1911	
10	88.9	185.64	141980	207.4	2.39	431.47	1.495	0.1911	
11	101.6	191.37	131727	203.0	2.26	441.57	1.546	0.1911	
12	114.3	197.18	122762	198.9	2.15	450.63	1.594	0.1911	
13	127	203.08	114823	195.2	2.05	459.01	1.639	0.1911	
14	139.7	209.07	107723	191.6	1.96	466.58	1.682	0.1911	
15	152.4	215.14	101,325	188.3	1.87	473.71	1.722	0.1911	Exit

Table 1: Isentropic Relationship Values

As shown in the table, the area decreases to the throat, then increases again, characteristic of a CD nozzle. The pressure decreases along the nozzle but has a pronounced dip at the throat, as expected from theory. The same results for the temperature as well, decreasing throughout the nozzle, but has a large dip at the throat. Density follows the same trend. The velocity is inversely proportional to the three previous properties; it increases throughout the nozzle, but increases sharply at the throat. The Mach number increases throughout the nozzle, while the mass flow rate remains constant, as expected.

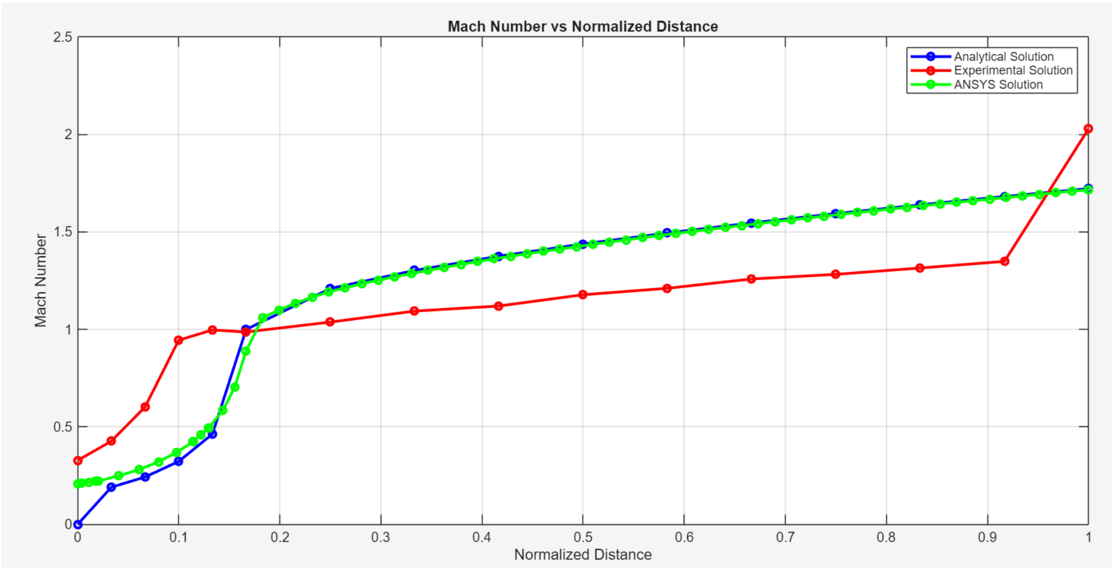

Figure 4: Mach Number vs. Normalized Distance

Figure 4 compares the experimental, analytical, and CFD Mach numbers as a function of the normalized nozzle distance. The analytical solution increases from subsonic to sonic at the throat and then continues to supersonic speeds past the throat, as expected for a CD nozzle. The experimental results show a similar trend, but have clear deviations. In the experimental results, the flow accelerates more rapidly upstream of the throat, likely due to sensitivity issues with the pressure transducers or uncertainty in the pitot tube. The experimental results are consistently lower than the analytical and CFD results in the diverging region, possibly due to non-ideal effects such as friction or other flow-related issues. The analytical results assume ideal, isentropic flow with no viscous losses or heat transfer throughout the nozzle. The CFD results show a very strong agreement with the analytical solution through most of the nozzle. As mentioned previously, the ANSYS model was assumed to be inviscid and ideal, which is why the analytical and CFD models align so closely. This close agreement indicates that the CFD model is correctly capturing compressible flow behavior, although the inviscid assumption limits its ability to represent experimentally observed flow losses fully. Minor deviations may be caused by rounding errors. At the exit, the experimental Mach number increases sharply and exceeds the analytical value. This sharp increase is likely nonphysical and may be due to measurement error or noise in the Pitot tube data near the nozzle exit. Overall, the comparison shows that the trends correspond, even though the experimental flow may not have been ideal.

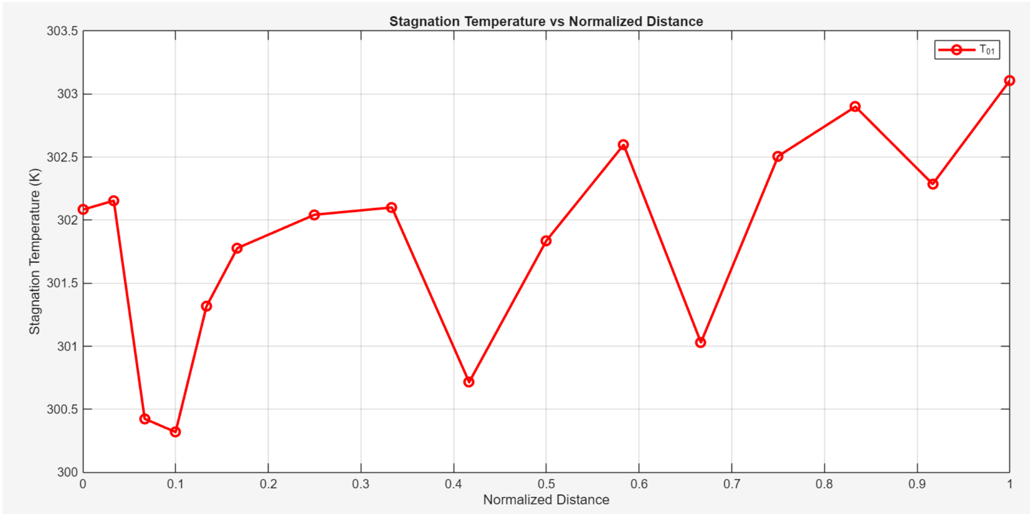

Figure 5: Temperature vs. Normalized distance

Figure 5 shows the stagnation temperature T_0, plotted against the normalized nozzle distance. Theoretically, stagnation temperature should remain constant in an isentropic flow since no heat transfer or work interactions occur. T_0 remains nearly constant, with only minor fluctuations, ranging from approximately 300 K to 303 K, which is consistent with the theory. These fluctuations can be due to measurement errors or non-ideal flow behavior, which can cause slight deviations in temperature. Overall, the temperature remains nearly constant, following the isentropic trend and aligning with our theoretical predictions. This figure was intended to compare T_{01} and T_{02}; however, the thermocouple attached to the Pitot tube produced unreliable readings, so those data were excluded from the analysis. As a result, only reservoir stagnation temperature (T_{01}) is presented to demonstrate the expected trend.

Discussion

The results from the analytical, experimental, and CFD calculations demonstrate the expected behavior of compressible flow through a CD nozzle. According to isentropic flow theory, a CD nozzle accelerates subsonic flow through the converging section until it reaches M=1 at the throat, then diverges to accelerate the flow to supersonic speeds. This is confirmed in Figure 4, where all three models show the Mach number increasing along the nozzle centerline. In the experimental model, the results follow the same trend as the analytical and CFD results, although with some clear deviations. These deviations can arise from non-ideal flow characteristics, such as viscous effects or boundary conditions, that are not accounted for in isentropic flow theory. The CFD results align very closely with the analytical results model, which show that the nozzle behaves as expected when simulated under ideal conditions. Minor deviations occur in the converging region, which may be caused by measurement uncertainty or rounding errors.

Overall, the experimental results and CFD simulation show strong agreement with the predictions of isentropic flow theory. The observed trends in Mach number, pressure, and temperature confirm that a converging nozzle accelerates subsonic flow and reaches the choked condition at the nozzle throat when the Mach number approaches one. Minor discrepancies between theory and experiment are expected due to measurement limitations and non-ideal flow effects. These results demonstrate that while isentropic theory accurately predicts overall flow behavior, real experimental flows are affected by viscous effects, instrumentation limitations, and flow disturbances, which introduce measurable deviations.

Conclusion

This experiment analyzed compressible flow through a CD nozzle under choked conditions. The analytical results showed the expected behavior, with the flow accelerating from subsonic to sonic in the converging section leading to the throat, and then from sonic to supersonic in the diverging region. The experimental results were similar, but with some deviations due to viscous effects, sensitivity, boundary conditions, and other non-ideal conditions. Stagnation temperatures also remained nearly constant. Overall, the results agree with compressible-flow theory and demonstrate the key characteristics of choked-flow conditions.

Appendix I

```
clear; clc; close all
R = 287;
g = 1.4;
T0 = 300;
%% Converging areas
r = 0.9842/2;
mC = (0.2794875 - r)/1;
xC = [0 0.25 0.5 0.75 1]';
AreaC = pi*(r + mC*xC).^2;
%% Diverging areas
r = 0.2794875;
mD = (0.3258 - r)/5;
xD = (1.5:0.5:6)';
AreaD = pi*(r + mD*(xD - 1)).^2;
%% Combine area and position
x = 25.4*[xC; xD];
A_in2 = [AreaC; AreaD];
Astar = AreaC(5);
%% Analytical Mach number
MAnalytical = zeros(15,1);
areaMach = @(M) (1./M).*((2/(g+1))*(1 +
((g-1)/2)*M.^2)).^((g+1)/(2*(g-1)));
for i = 1:15
   A_ratio = A_in2(i)/Astar;
   if i == 5
       MAnalytical(i) = 1;
   elseif i < 5
       MAnalytical(i) = fzero(@(M) areaMach(M) - A_ratio, 0.5);
   else
       MAnalytical(i) = fzero(@(M) areaMach(M) - A_ratio, 1.5);
   end
end
%% Pressure calculations
pe = 101325;
p0 = pe*(1 + ((g-1)/2)*MAnalytical(end)^2)^(g/(g-1));
p = p0*(1 + ((g-1)/2)*MAnalytical.^2).^(-g/(g-1));
%% Temperature calculations
T = T0*(1 + ((g-1)/2)*MAnalytical.^2).^(-1);
%% Density calculations
```

```
rho = p./(R*T);
%% Velocity calculations
a = sqrt(g*R*T);
v = MAnalytical.*a;
%% Mass flow calculations
A_m2 = 0.00064516*A_in2;
A_mm2 = A_m2 * 1e6;
mDot = rho.*A_m2.*v;
%% Notes column
Notes = strings(15,1);
Notes(1) = "Inlet";
Notes(5) = "Throat";
Notes(15) = "Outlet";
%% Round values for table
position_mm = x;
Area_mm2 = round(A_mm2,2);
Mach = round(MAnalytical,3);
Pressure_Pa = round(p,0);
Temperature_K = round(T,1);
Density_kg_m3 = round(rho,4);
Velocity_m_s = round(v,1);
MassFlow_kg_s = round(mDot,4);
%% Create table
results =
table(position_mm,Area_mm2,Mach,Pressure_Pa,Temperature_K,Density_kg_m3,Ve
locity_m_s,MassFlow_kg_s,Notes,...
   'VariableNames',
{'Position_mm','Area_mm2','Mach','Pressure_Pa','Temperature_K', ...
   'Density_kg_m3','Velocity_m_s','MassFlow_kg_s', 'Notes'});
% Display table
disp(results)
writetable(results, 'Nozzle_Flow_Results.xlsx');
disp('Table saved as Nozzle_Flow_Results.xlsx')
%% Plots
data = readmatrix("203_B1.xlsx");
CFD = readmatrix('DataNozz.xlsx');
currentLocation = data(:,3);
p01 = flip(6895*data(:,4));
p02 = flip(6895*data(:,5));
ps = flip(6895*data(:,6));
T01 = flip(273.15 + data(:,7));
normDist = flip((currentLocation-1)./6);
MAnalytical =[0; MAnalytical];
MAnalytical(6) = 1;
M = sqrt((2/.4)*(((ps./p02).^(-.4/1.4))-1));
CFDM = CFD(:,9);
CFDNorm = CFD(:,4);
```

```
figure(1)
plot(normDist,MAnalytical,'bo-', 'Linewidth', 2, 'MarkerSize', 5)
hold on
plot(normDist,M,'ro-','LineWidth', 2, 'MarkerSize', 5);
hold on
plot(CFDNorm,CFDM,'go-','LineWidth', 2, 'MarkerSize', 5)
xlabel('Normalized Distance')
ylabel('Mach Number')
title('Mach Number vs Normalized Distance')
grid on
legend('Analytical Solution','Experimental Solution','ANSYS Solution')
figure(2)
plot(normDist(:),T01,'ro-', 'LineWidth', 2, 'MarkerSize', 7)
grid on
xlabel('Normalized Distance')
ylabel('Stagnation Temperature (K)')
title('Stagnation Temperature vs Normalized Distance')
legend('T_{01}')
```

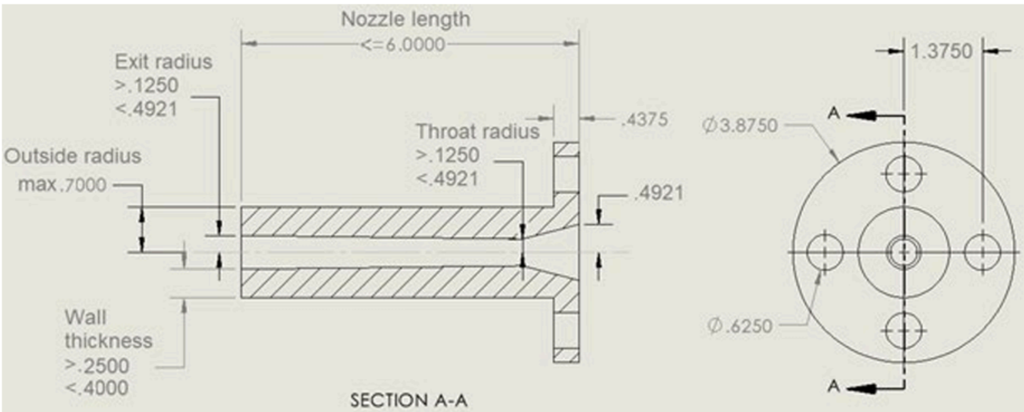

Figure 6: Constraints on the CD nozzle geometry (unit: inch)

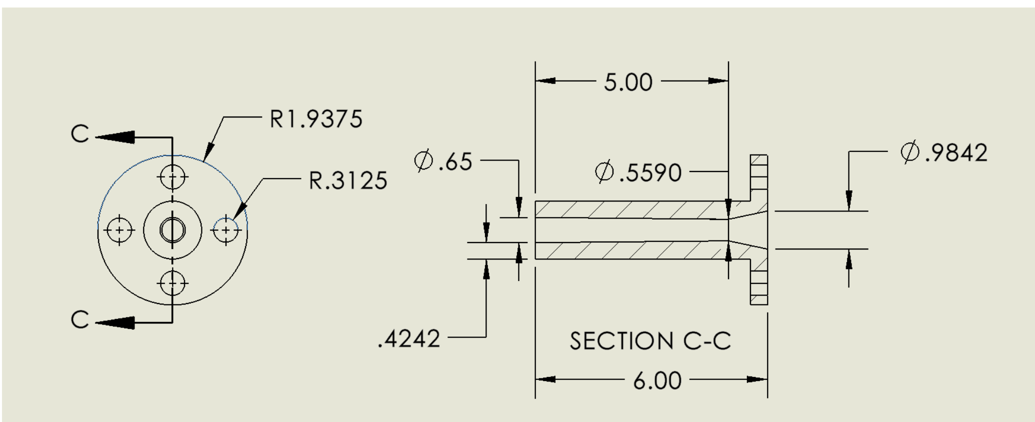

Figure 7: Nozzle Design Specifications

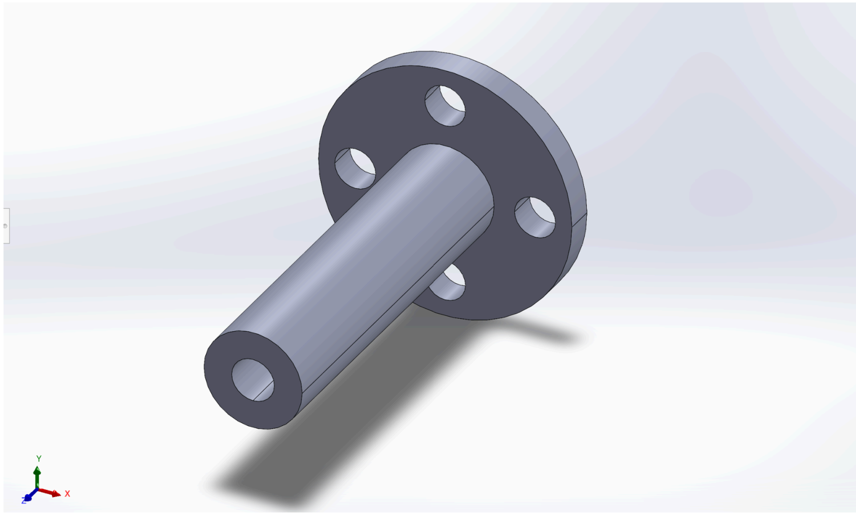

Figure 8: SolidWorks model of M = 1.72 nozzle

Appendix II

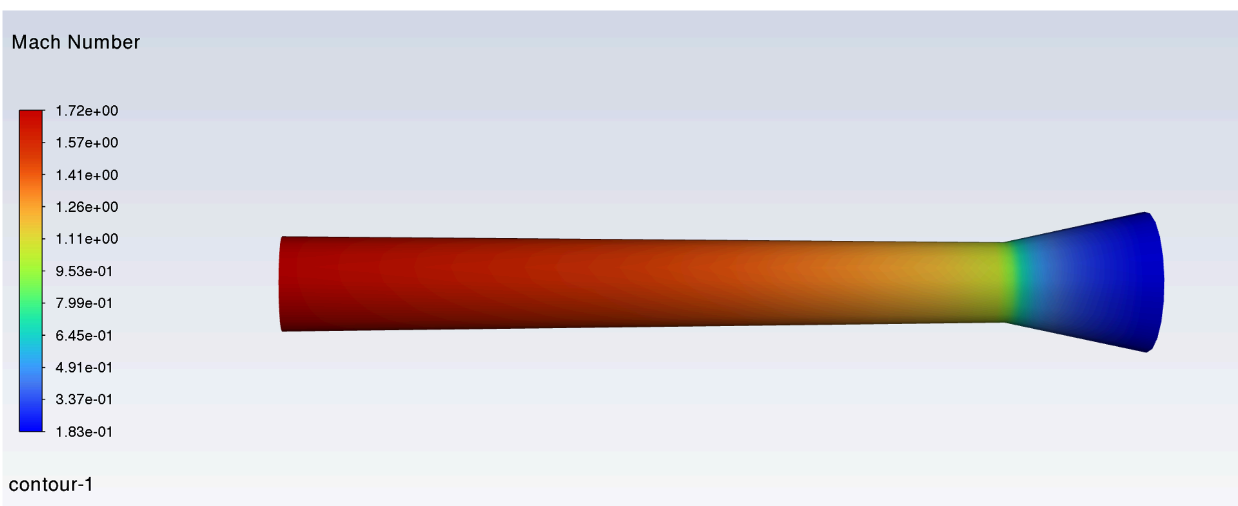

Figure 9: Contour map from CFD analysis showing the Mach number in relation to nozzle geometry.

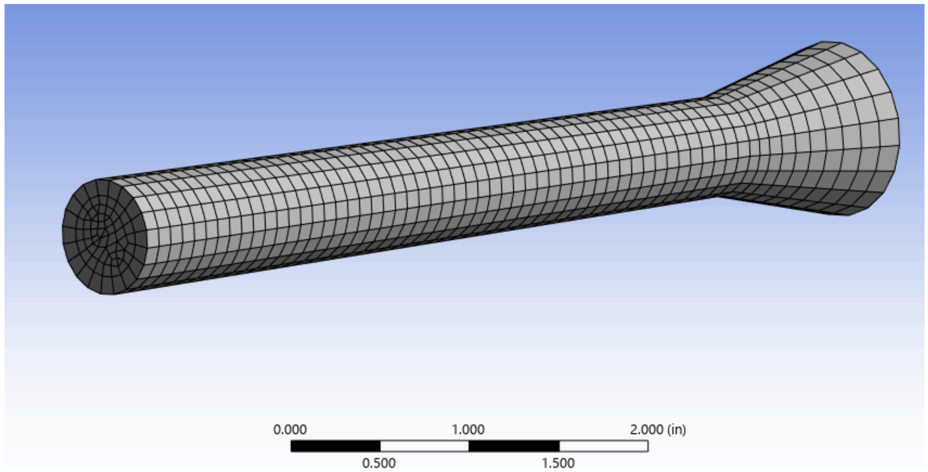

Figure 10: Image of Computational Mesh Used in CFD Analysis